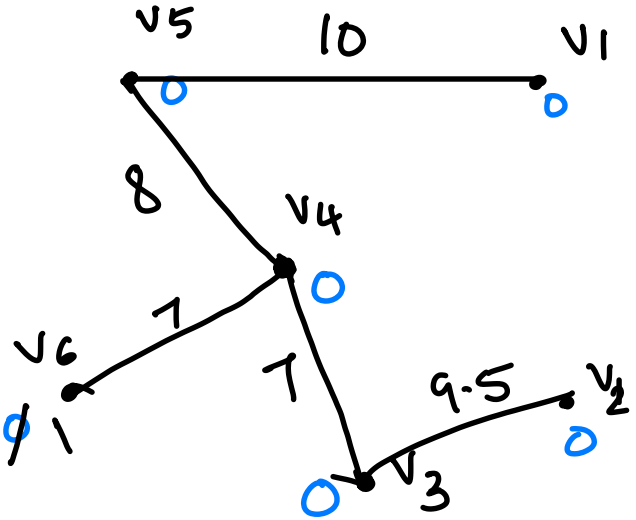
Operations Involved:

Make-Set (x)

1. $P[x] \leftarrow x$
2. $rank[x] \leftarrow 0$

$P[x] \leftarrow$ parent of node x

Each vertex parent is itself and rank 0



Sorted :

all parent = v_6 = root.

E

✓	✓	✓	X	✓	✓	X	X	X		
~~v_4v_6~~	v_3v_4	v_4v_5	v_5v_6	v_2v_3	v_5v_1	v_1v_2	v_4v_1	v_3v_6	v_3v_1	v_6v_1	v_6v_2
7	7	8	9	9.5	10	10	11	12	16	17	19.5

$(u,v) = (v_4, v_6) \in E$ ✓

Find-set(v_4) = v_4 ($\because P[v_4] = v_4$)

Find-set(v_6) = v_6 ($\because P[v_6] = v_6$)

$v_4 \neq v_6$

$A \leftarrow \phi \cup \{(v_4, v_6)\} = \{(v_4, v_6)\}$

Union (v_4, v_6)

$\|$

Link(Find-Set(v_4), Find-Set(v_6))

$=$ Link(v_4, v_6)

ie add an edge and increase rank or update parent.

Did p[v_4] update?

$0 = \text{rank}[v_4] = \text{rank}[v_6] = 0$

$\Rightarrow \text{rank}[v_6] = 0 + 1 = 1$

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and add the edge $v_4 v_6$.

$(u,v) = (v_3, v_4) \in E$

Find_set $(v_4) = v_6 \neq v_3 =$ Find_set (v_3)

$A \leftarrow A \cup \{(v_3, v_4)\}$

$= \{(v_4, v_6), (v_3, v_4)\}$

Union (v_3, v_4)

↓

Link(Find-set(v_3), Find_set(v_4))

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$= \text{Link}(v_3, v_6)$

$0 = \text{rank}[v_3] \neq \text{rank}[v_6] = 1.$

$P[v_3] \leftarrow y = v_6$

Add $v_3 v_4$ edge.

Note:

Find_set (v_4):

$v_4 = x \neq P[x] = v_6$

$P[x] \leftarrow$ Find_set (v_6)

Find_Set (v_6).

$v_6 = x = P[x] = v_6$

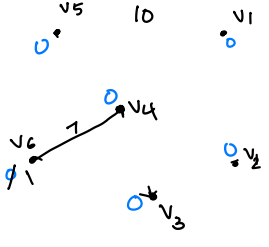
$\therefore \quad P[x] = v_6.$

Find_set (v_4) $= v_6$.

Find-set (x) → see in graph.

1. If $x \neq P[x]$
2. then $P[x] \leftarrow$ Find-set ($P[x]$)
3. return $P[x]$

Graph:



$(u, v) = (v_3, v_4)$

Find_set(v_5) = Find_set(v_4)

$\|$ $\|$

v_5 v_6.

Union(v_5, v_4)

link(v_5, v_6)

$0 =$ rank$[v_5] \not>$ rank$[v_6] = 1$

$P[v_5] \leftarrow v_6$.

$(u, v) = (v_6, v_5)$

Find_set(v_6) = Find_set(v_6)

$\|$ $\|$

v_6 v_6.

(Not considered)

$(u, v) = (v_3, v_2)$

$V_6 = \text{Find_set}(V_3) \neq \text{Find_set}(V_2) = V_2$

$A \leftarrow \{(V_4, V_6), (V_3, V_4), (V_5, V_4), (V_3, V_2)\}$

$\text{Union}(V_3, V_2)$

$\parallel$

$\text{Link}(\text{Find_set}(V_3), \text{Find_set}(V_2))$

$= \text{Link}(V_6, V_2)$

$1 = \text{rank}[v_0] > \text{rank}[v_2] = 0$

$P[v_2] \leftarrow v_6$.

$(u, v) = (v_1, v_5) \in E$.

$v_1 = \text{Find-set}(v_1) \neq \text{Find-set}(v_5) = v_6$

$A \leftarrow \{(v_4, v_6), (v_3, v_4), (v_5, v_4), (v_3, v_2),$

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$(v_{11} v_5)\}$

$\text{Union}(v_{11} v_5) = \text{Link}(v_{11}, v_6)$

$D = \text{rank}[v_1] \not> \text{rank}[v_6] = 1$

$\text{rank}[v_1] \leftarrow v_6$

$(u, v) = (v_1, v_2) \in E$.

$v_6 = \text{Find_set}(v_1) = \text{Find_set}(v_2) = v_6$.

Similar for rest of the edges.

(Stops)

Analysis:

Functional Iteration Notation

$$f : \mathbb{N} \to \mathbb{R}$$

$$f^{(i)}(n) = \begin{cases} n & \text{if } i = 0 \\ f\left(f^{(i-1)}(n)\right) & \text{if } i > 0 . \end{cases}$$

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eg: $f(n) = 2n$

$$f^{(i)}(n) = \begin{cases} n & i = 0 \\ f(f^{i-1}(n)) & i > 0 \end{cases}$$

$i > 0$.

$i = 1$.

$$f^{(1)}(n) = f\left(f^{(0)}(n)\right) = f(n) = 2n$$

$i=2$

$$f^{(2)}(n) = f(f^{(1)}(n)) = f(2n) = 2(2n)$$

$$= 2^2 n$$

$$f(n) = n = 2^0 n$$

$f^{(i)}(n) = 2^i n$ $\quad i \geq 0$

← (A guess).

Prove $f^{(i)}(n) = 2^i n$ by induction.

Let $i = 0$.

$f(n) = n = 2^0 n$.

Assume the result is true for $i=k$.

$$f^{(k)}(x) = 2^k x$$

We shall prove the result for $i=k+1$

$$f^{k+1}(n) = f(f^k(n))$$

$$= f(2^k n)$$ (By induction hypothesis)

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$$= 2(2^k n)$$

$$= 2^{k+1} n.$$

$\square$ s

QED

A very quickly growing function with very slowly growing inverse:

For integers $k \geq 0$; $j \geq 1$

$$A_k = \begin{cases} j+1 & \text{if } k=0 \\ A_{k-1}^{(j+1)}(j) & \text{if } k \geq 1. \end{cases} \quad \text{---(1)}$$

k is called level of function A.

$$A_{k-1}^{(0)}(j) = j \quad - (2)$$

(Initialization)

$$A_{k-1}^{(i)}(j) = A_{k-1}\left(A_{k-1}^{i-1}(j)\right) - (3)$$

(Ex proxy Rule)

Results :

I) For any integer $j \geq 1$.

(i) $A^{(i)}_{(0)}(j) = j + i$

(ii) $A_1(j) = 2j + 1$

Proof:

(Prove by induction)

T.P: $A_0^{(i)}(j) = j+1$.

③ $\rightarrow A_{k-1}^{(i)}(j) = A_{k-1}\left(A_{k-1}^{i-1}(j)\right)$

Put $k=1$. in ③.

$$A_0^{(i)}(j) = A_0\left(A_0^{i-1}(j)\right)$$

We prove (i) by induction on i.

Take $i=1$

$$A_0^{1}(j) = A_0\left(A_0^{(0)}(j)\right) \qquad (\text{put } k=1 \text{ in } 3)$$

$$= A_0(j) \qquad \text{put } k=1 \text{ in } (2)$$

$$= j+1.$$

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$\therefore$ The result is true for $i=1$.

Assume the result is true for $i=t$.

i.e $A_0^t(j) = j+t$.

To prove the result for $i=t+1$.

$$A_0^{t+1}(j) = A_0\left(A_0^t(j)\right)$$

By ③ with $k=1$.

$$= A_0(j+t)$$

(By induction hypothesis)

$$= j+t+1$$

(By ①) □

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(ii) $A_1^{(j)}(j) = A_0^{j+1}(j)$ (by ①).

$= j + (j+1)$ (By (i))

$= 2j + 1$

$\square$